SMOOTHNESS APPEARS AT A FINITE STAGE IN THE DESCENT SITUATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1737-1773, following the statement at 1726-1735.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Smooth is etale-local on the base. Morphisms of Spaces, Definition spaces-morphisms-definition-smooth (cross-chapter, cited at ~1766-1769). X_i -> Y_i smooth iff U_i -> V_i smooth; likewise for X -> Y.
- Scheme-level descent of smoothness. Limits, Lemma limits-lemma-descend-smooth (cross-chapter, cited at ~1771-1772). The engine for U_i -> V_i.

Source reference. Original reference 0CN2: focused-seeds/geometry/S9-spaces-limits-descend-smooth/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1666–1666. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03ZC: focused-seeds/geometry/S9-spaces-limits-descend-smooth/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *With notation and assumptions as in Situation 084W. If*

(1) *f is smooth,*
(2) *f_0 is locally of finite presentation,*

then f_i is smooth for some $i \geq 0$.

Proof. Choose an affine scheme V_0 and a surjective étale morphism $V_0 \to Y_0$. Choose an affine scheme U_0 and a surjective étale morphism $U_0 \to V_0 \times_{Y_0} X_0$.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Diagram

$$\begin{array}{ccc} U_0 & \longrightarrow & V_0 \\ \downarrow & & \downarrow \\ X_0 & \longrightarrow & Y_0 \end{array}$$

The vertical arrows are surjective and étale by construction. We can base change this diagram to B_i or B to get

$$\begin{array}{ccc} U_i & \longrightarrow & V_i \\ \downarrow & & \downarrow \\ X_i & \longrightarrow & Y_i \end{array} \quad \text{and} \quad \begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

Note that U_i, V_i, U, V are affine schemes, the vertical morphisms are surjective étale, and the limit of the morphisms $U_i \to V_i$ is $U \to V$. Recall that $X_i \to Y_i$ is smooth if and only if $U_i \to V_i$ is smooth and similarly $X \to Y$ is smooth if and only if $U \to V$ is smooth (Morphisms of Spaces, Definition 03ZC). Since f_0 is locally of finite presentation, so is the morphism $U_0 \to V_0$. Hence the lemma follows from Limits, Lemma 0CN2. $\square$